

Short Questions

Q. Where Application resources are stored?

⇒ Application resources are stored under the "res/folder" of project hierarchy. In this folder, each of the available resource types can have a sub-folder containing its resources.

Q. List the primary resource types?

⇒ There are seven primary resource types that have different folders:

- Simple values
- Drawables
- layouts
- animations
- XML styles
- raw resources.

⇒ When application is built, these resources will be compiled as

efficiently as possible and included in the application package.

Q. What is R class file?

⇒ The R class gets code generated by the IDE and build tools based on current roster of resources.

⇒ R classes are generated for every module of build and the module specific R classes include references to all resources of its transitive dependencies.

⇒ R is a static class that list all resources.

⇒ R class file contains the references to each of the resources include in project.

⇒ it is usually defined in XML but all available in res folder.

⇒ R class is located under:
app\build\generated\source\res

⇒ Each resource gets a corresponding R value based on the type of resource (R.layout, R.id, R.string, etc) and the name of the resource (R.

Q. Caution while keeping resource file name?

⇒ In all cases, the resource file names should contain only lower case letters, numbers, and the period(.) and underscore(-) symbols.

Q. How to create simple values?

⇒ Simple values include:

- Strings
- Colors
- dimensions
- String or integer arrays

⇒ All simple values are stored within XML files in the res/values folder.

Within each XML file, to indicate the type of value being stored using tags are as follows:

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<resources>
```

```
<string name="app-name">To Do List</string>
```

```
<color name="app-background">#FF0000</color>
```

```
</color>
```

```
<dimen name="default-border">5px</dimen>
```

```
</dimen>
```

```
<array name="string-array">  
<item>Item 1</item>
```

```
:
```

```
</array>
```

```
<array name="integer-array">  
<item>Item 1</item>
```

```
:
```

```
</array>
```

```
</resources>
```

Strings:

Externalizing strings helps maintain consistency within application and makes it much easier to create localized versions.

String resources are specified by using the string tag.

Colors:

⇒ Color tag is used to define a new color resource.

⇒ Notations are:

o #RGB

o #RRGGBB

o #ARGB

o #ARRGGBB

Dimensions:

⇒ Dimensions are most commonly referenced within style and layout resources.

⇒ They are useful for creating layout constants such as borders and font heights.

⇒ To specify a dimension resource, use the `dimen` tag, specifying the dimension value followed by an identifier describing the scale of dimension:

- `px` Screen pixels
- `in` Physical inches
- `pt` physical points
- `mm` physical millimeters
- `dp` Density independent pixels
- `sp` Scale independent pixels

⇒ To specify dimension values for a large font size and the standard border:

```
◦ <dimen name="standard-border">
    50px </dimen>
```

```
◦ <dimen name="large-font-size">
    16sp </dimen>
```

Styles and themes:-

⇒ Style resources maintain a consistent look and feel by specifying the attribute values used by Views.

⇒ The most common use of themes and styles is to store the colors and fonts for an application.

⇒ It is easy to change the appearance of application by simply specifying a different style as the theme in project manifest.

How to create style?

⇒ To create a style, use a `style` tag that includes a name attribute, and contains one or more item tags.

⇒ Each item tag should include a name attribute used to specify the attribute being defined.

<resources>

<style name="StyleName">

<item name="attributeName">

value </item>

</style>

</resources>

Q. What is drawable folder?

⇒ The drawable folder includes resources like bitmaps and NinePatch (stretchable PNG) images. They are stored as individual files in the res/drawable folder.

⇒ The resource identifier for a bitmap resource is the lowercase filename without an extension.

⇒ The preferred format for a bitmap resource is PNG, although JPG and GIF files are also supported.

⇒ A NinePatch is a variation of a PNG image that uses a one-pixel border to define the area

of the image that can be stretched if the image is enlarged.

⇒ NinePatches are a powerful technique for creating images for the background of Views or Activities that may have a variable size.

e.g. Android uses NinePatches for creating button backgrounds.

Q. Describe the types of Animations supported by android?

⇒ Android supports two types of animation.

- Tweened Animation
- Frame-by-frame Animation

Tweened Animation:

⇒ Tweened animation can be used to rotate, move, stretch and fade a View.

⇒ Each Tweened animation
Stored in a separate XML file
in the project res/anim folder.

⇒ An animation can be defined
for changes in:

- alpha (fading)
- scale (scaling)
- translate (moving)
- rotate (rotating)

Frame-by-Frame Animation:-

⇒ Create a sequence of drawables,
each of which will be displayed
for a specified duration, on the
background of a View.

⇒ Because frame-by-frame animations
represent animated drawables,
they are stored in the res/drawable
folder, rather than with the Tweened
animations, and use their filenames
as their resource ids.

Q. How you debug your Apps?

⇒ The following ways are:

- Using log
- Using Logcat
- Using debug mode
- Attach a debugger
- Using break points
- Using Toast